An introduction to Quantum Computing using Qiskit

Course ID 5841

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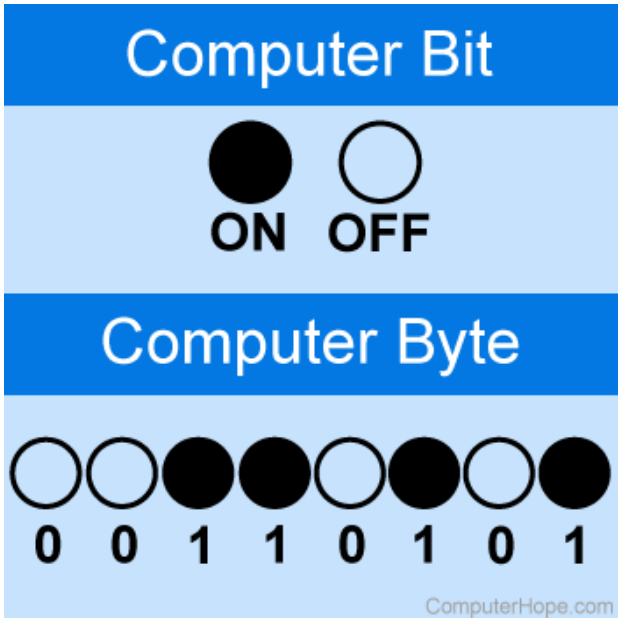
*(reference slides created by: **Gayathree M. Vinod**)*

Classical computation:

- fundamental unit of classical information – bit



state – either 0 or 1

Qubit:

- **Two possible states** - $|0\rangle$ and $|1\rangle$
  - computational basis states

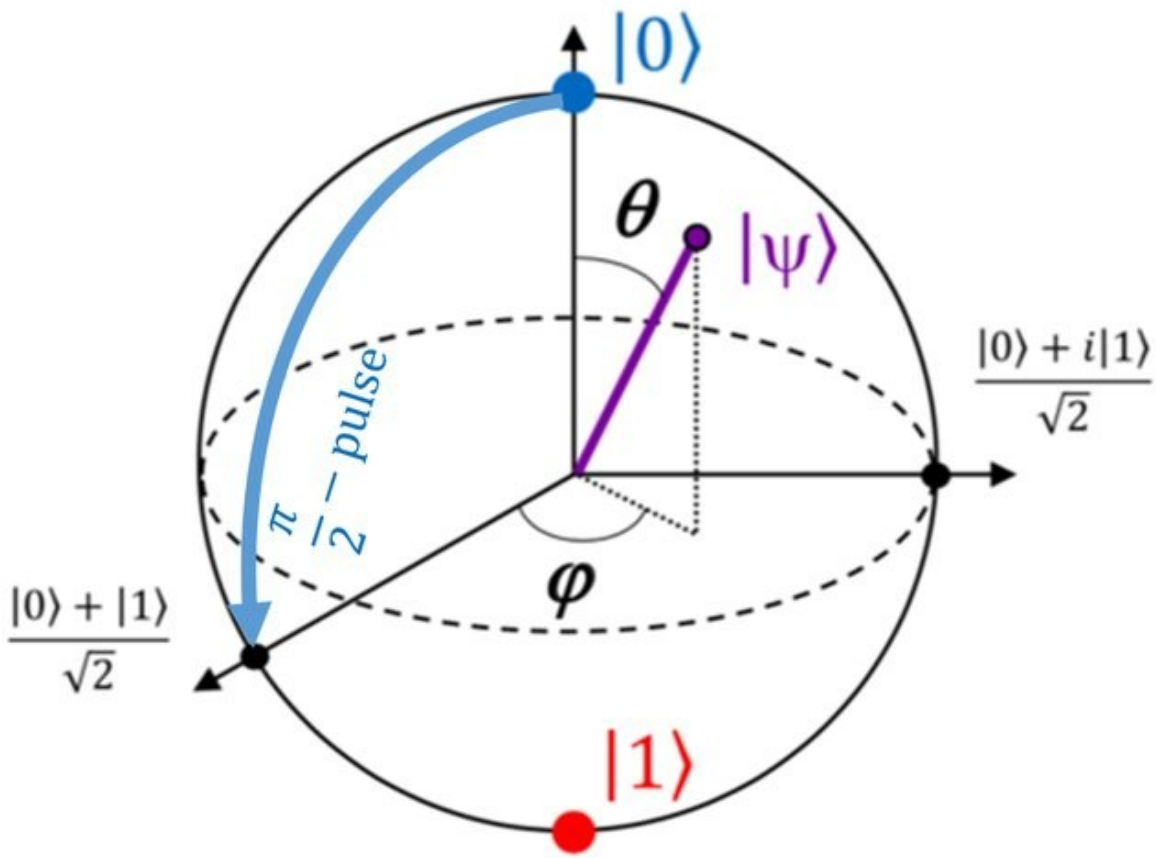
- `$|\ \rangle$' $\rightarrow$ Dirac notation
- difference from bits – a qubit can form ***linear combinations* of states** – ***'superpositions'***

$$|\boldsymbol{\psi}\rangle = \boldsymbol{\alpha}|\mathbf{0}\rangle + \boldsymbol{\beta}|\mathbf{1}\rangle$$

(α, β $\rightarrow$ complex numbers)

$$|\alpha|^2 + |\beta|^2 = 1$$

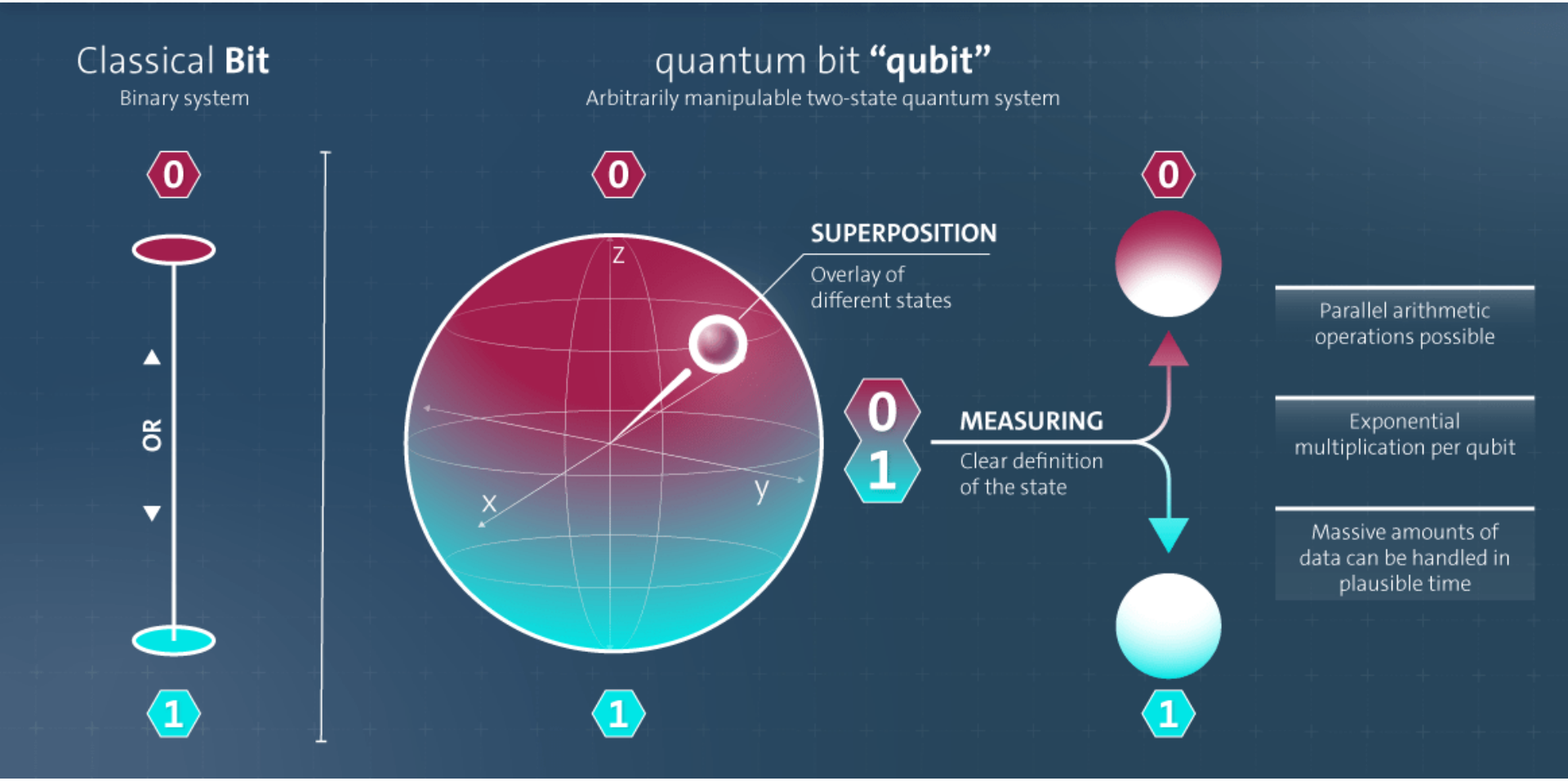
Bloch sphere representation of a qubit:



$$|\psi\rangle = cos\frac{\theta}{2}|0\rangle + e^{i\varphi}sin\frac{\theta}{2}|1\rangle$$

https://doi.org/10.48550/arXiv.1908.02656

Bloch Sphere:



https://www.volkswagenag.com/en/news/stories/2019/11/where-is-the-electron-and-how-many-of-them.html

Quantum state vectors:

- represented by a **column vector**, $v = \begin{pmatrix} \alpha_1 \\ \vdots \\ \vdots \\ \alpha_n \end{pmatrix}$
- Properties:
  - The **entries** are **complex** numbers
  - **Sum of the absolute values squared** of the entries is **1**

The ***Euclidean norm*** of a column vector:

$$||v|| = \sqrt{\sum_{k=1}^{n} |\alpha_k|^2}$$

=> **unit vectors** w.r.t. the Euclidean norm

Examples:

- $|\mathbf{0}\rangle = \begin{pmatrix}\mathbf{1}\\ \mathbf{0}\end{pmatrix}$ **and** $|\mathbf{1}\rangle$ **=** $\begin{pmatrix}\mathbf{0}\\ \mathbf{1}\end{pmatrix}$

  - computational basis states

- $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix}$,

- $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{-1}{\sqrt{2}}\end{pmatrix}$

$\langle\psi| \rightarrow$ conjugate transpose of $|\psi\rangle$ ($\therefore$ row vector)

$\langle\psi|\psi\rangle \rightarrow$ inner product

Unitary operations:

- A square matrix U is ***unitary*** if

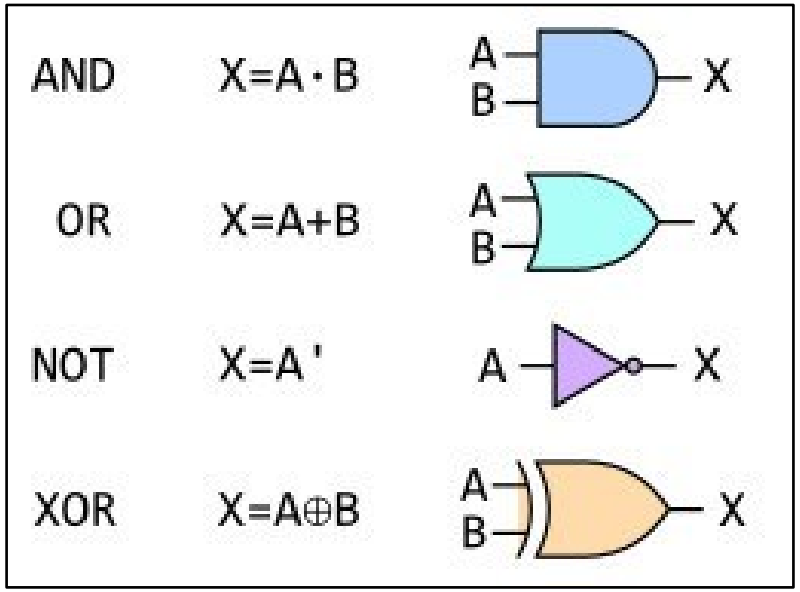
$$UU^{\dagger} = I$$
$$U^{\dagger}U = I$$

$U^{\dagger} \rightarrow$ conjugate transpose of U

- equivalent to the condition that multiplication by U **does not change the Euclidean norm** of any vector

Quantum state vector $\xrightarrow{mapped\ by\ U}$ **Quantum state vector**

Boolean gates:

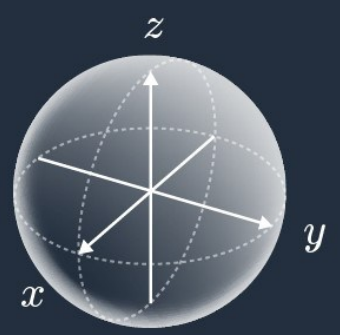
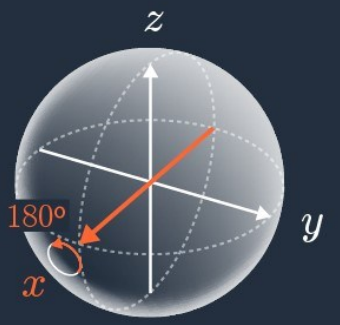


Truth tables

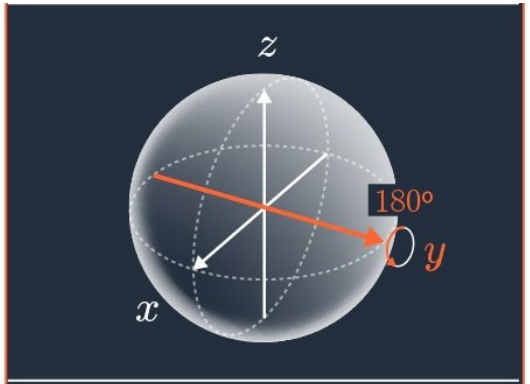
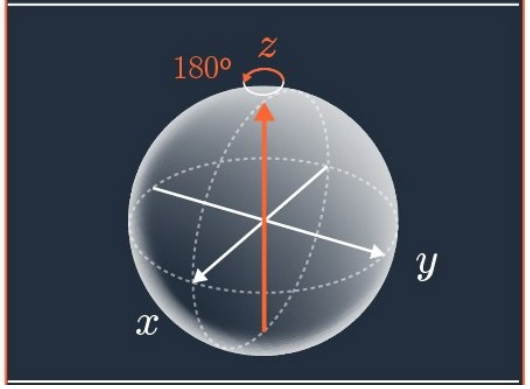
A	B	A·B	A+B	A'	A⊕B
0	0	0	0	1	0
0	1	0	1	1	1
1	0	0	1	0	1
1	1	1	1	0	0

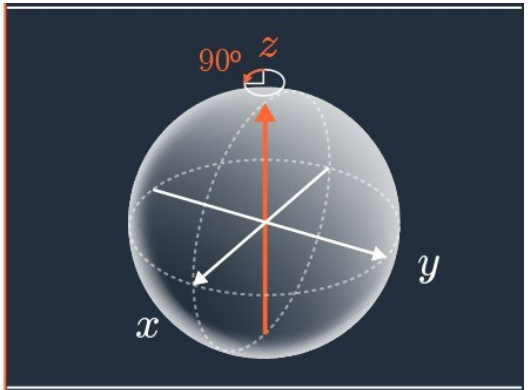
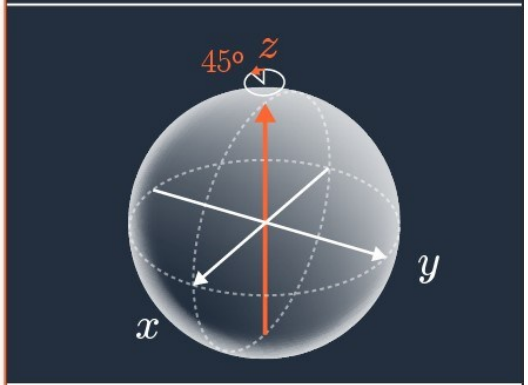
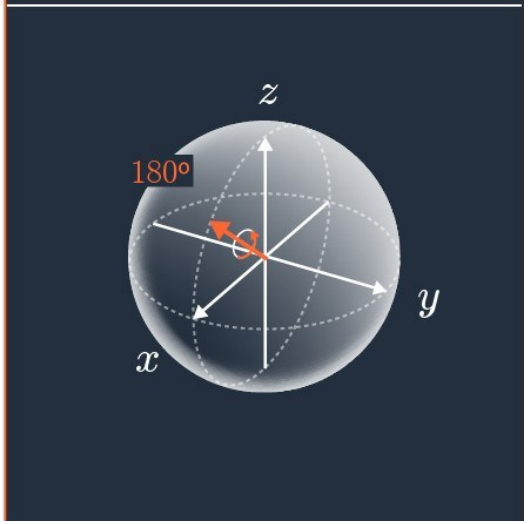
https://liucs.net/cs101s19/boolean.html

Single qubit gates:

GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE	BLOCH SPHERE
I Identity-gate: no rotation is performed.	$\boxed{I}$	$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	Input / Output $\lvert 0\rangle$ / $\lvert 0\rangle$ $\lvert 1\rangle$ / $\lvert 1\rangle$	
X gate: rotates the qubit state by π radians (180º) about the x-axis.	$\boxed{X}$	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Input / Output $\lvert 0\rangle$ / $\lvert 1\rangle$ $\lvert 1\rangle$ / $\lvert 0\rangle$	

https://github.com/att/quantum_workshop/blob/master/solutions/single-qubit-gates.ipynb

Y gate: rotates the qubit state by π radians ($180°$) about the y-axis.	$-\boxed{Y}-$	$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Input: $\lvert 0\rangle$, $\lvert 1\rangle$ Output: $i\,\lvert 1\rangle$, $-i\,\lvert 0\rangle$	
Z gate: rotates the qubit state by π radians ($180°$) about the z-axis.	$-\boxed{Z}-$	$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Input: $\lvert 0\rangle$, $\lvert 1\rangle$ Output: $\lvert 0\rangle$, $-\lvert 1\rangle$	

S gate: rotates the qubit state by $\frac{\pi}{2}$ radians ($90°$) about the z-axis.	S	$S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix}$	Input: $\lvert 0\rangle$, $\lvert 1\rangle$ Output: $\lvert 0\rangle$, $e^{i\frac{\pi}{2}}\lvert 1\rangle$	
T gate: rotates the qubit state by $\frac{\pi}{4}$ radians ($45°$) about the z-axis.	T	$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}$	Input: $\lvert 0\rangle$, $\lvert 1\rangle$ Output: $\lvert 0\rangle$, $e^{i\frac{\pi}{4}}\lvert 1\rangle$	
H gate: rotates the qubit state by π radians ($180°$) about an axis diagonal in the x-z plane. This is equivalent to an X-gate followed by a $\frac{\pi}{2}$ rotation about the y-axis.	H	$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Input: $\lvert 0\rangle$, $\lvert 1\rangle$ Output: $\frac{\lvert 0\rangle + \lvert 1\rangle}{\sqrt{2}}$, $\frac{\lvert 0\rangle - \lvert 1\rangle}{\sqrt{2}}$	

The Pauli matrices:

$$X \text{ or } \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$Y \text{ or } \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$Z \text{ or } \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma_i^2 = 1$$

Anticommutation relation:

$$\{\sigma_j, \sigma_k\} = 2\delta_{jk}$$

Commutation relation:

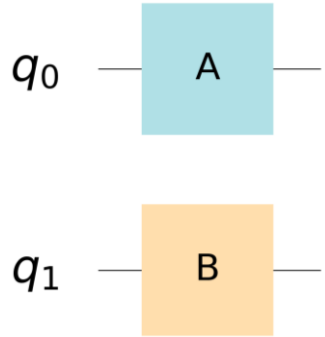
$$[\sigma_j, \sigma_k] = 2i\varepsilon_{jkl}\sigma_l$$

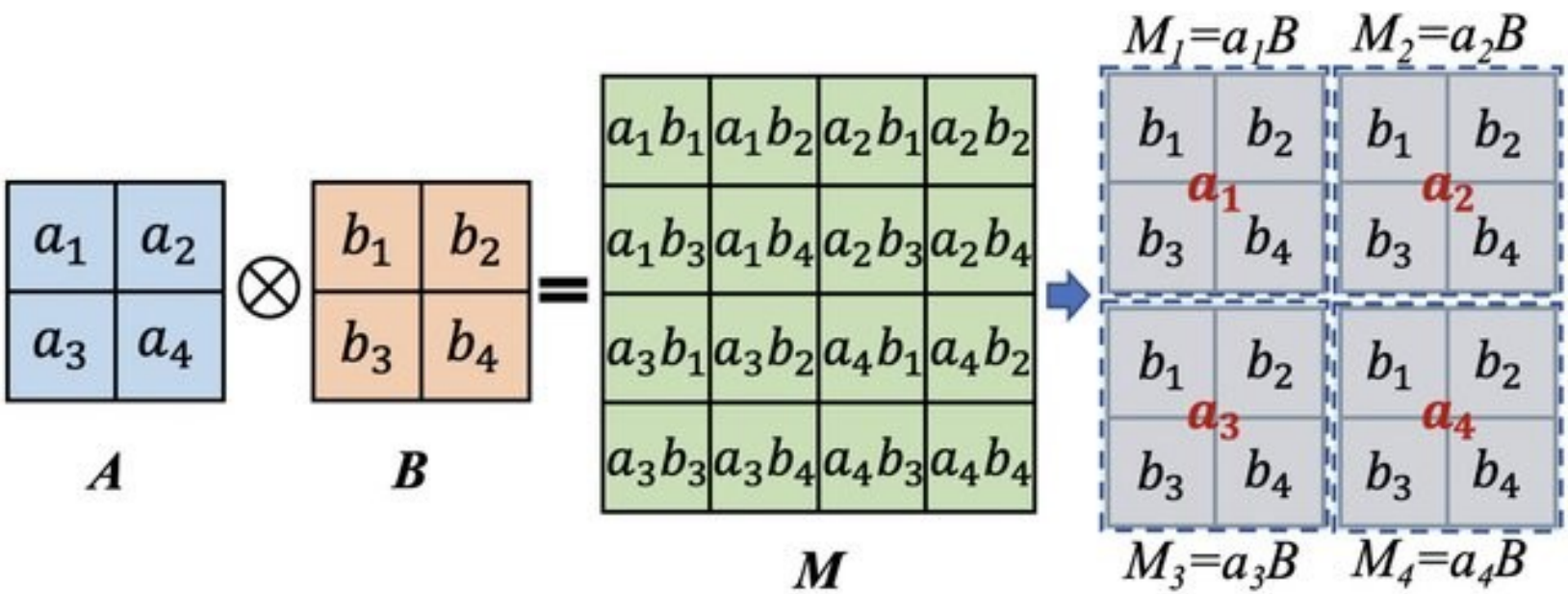
The Rotation operators:

$$R_x(\theta) \equiv e^{-i\theta X/2}$$
$$R_y(\theta) \equiv e^{-i\theta Y/2}$$
$$R_z(\theta) \equiv e^{-i\theta Z/2}$$

Multi qubit states:

- **Tensor product** $|ab\rangle = |a\rangle|b\rangle = |a\rangle \otimes |b\rangle$





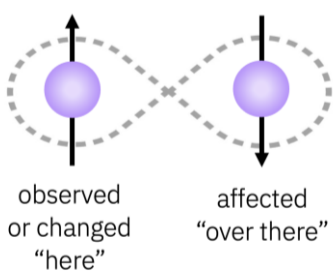
(2x2) (2x2) (2^2 x 2^2 = 4)

Song, Y., Sha, E.HM., Zhuge, Q. *et al.* Efficient algorithm for full-state quantum circuit simulation with DD compression while maintaining accuracy. *Quantum Inf Process* **22**, 413 (2023). https://doi.org/10.1007/s11128-023-04160-5

Entanglement:

- **Quantum entanglement:** occurs when two or more qubits are intrinsically linked, such that one cannot describe the state of any qubit in the entangled system independently.

- Entangled qubits are correlated more strongly than classical probability allows.
- Operating on or collapsing one qubit instantly influences another qubit in the entangled system (although it never violates relativity).



- Example:

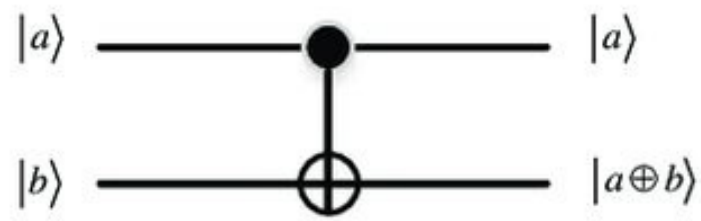
- $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

- If we measure one qubit to be $|0\rangle$ in the z-basis, the measurement collapses the two-qubit state, leaving the other qubit to be $|0\rangle$.

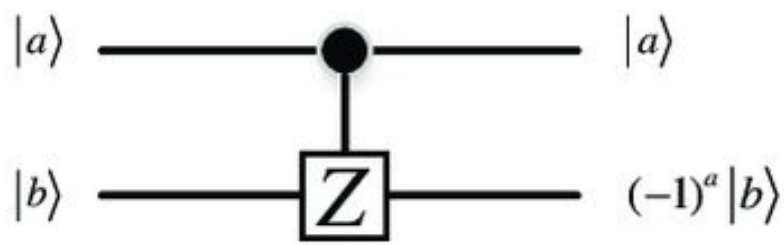
- Instead, if we measure in the x-basis and find one qubit to be $|+\rangle$, the other qubit collapses simultaneously to be $|+\rangle$.

- **Entangled states:**
  - *correlated* states
  - cannot be written as product states
  - (a **resource** for quantum computation)

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



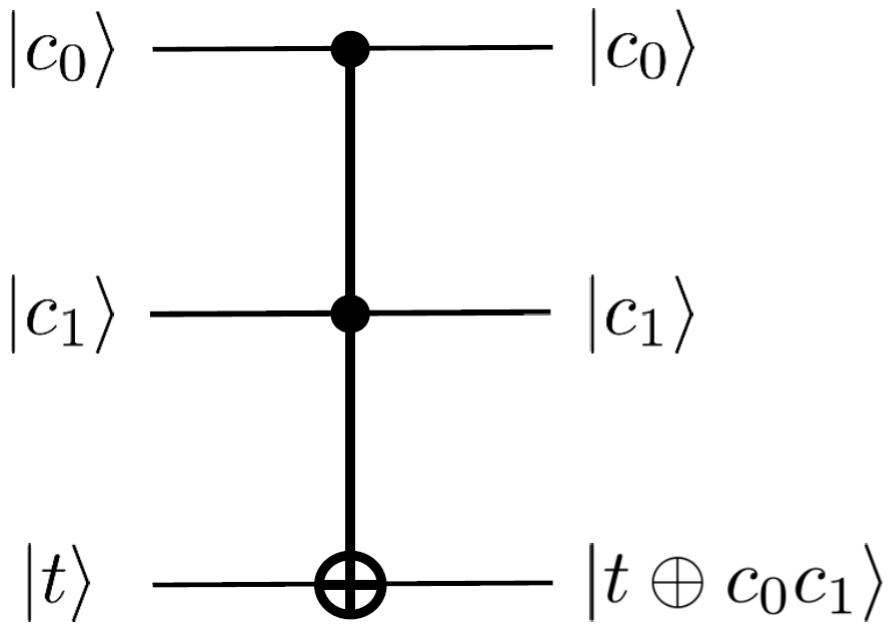
$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$



https://medium.com/@soumyojyotidatta/exploring-quantum-entanglement-bdfe18dea2a3

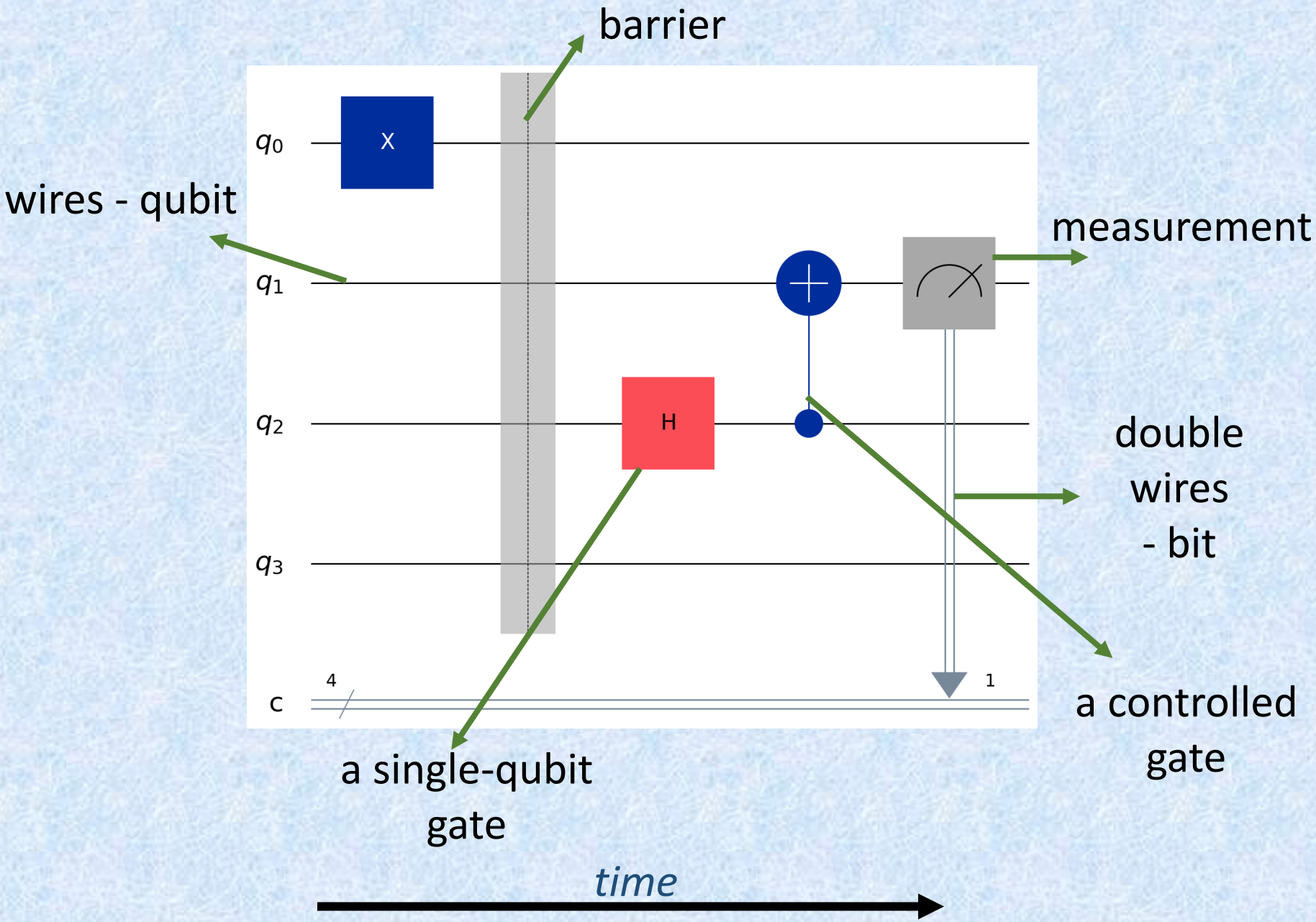
W. Liu, Y. Xu, W. Liu, H. Wang and Z. Lei, "Quantum Searchable Encryption for Cloud Data Based on Full-Blind Quantum Computation," in *IEEE Access*, vol. 7, pp. 186284-186295, 2019, doi: 10.1109/ACCESS.2019.2960592.

Toffoli gate:



$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Quantum circuit



Examples:

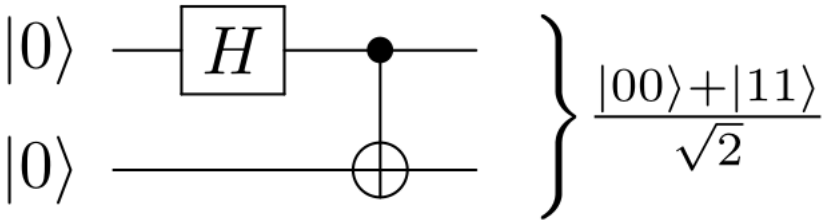
A. Bell states (maximally entangled)
(a resource for quantum computation)

$$|\beta_{00}\rangle \text{ or } |\phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$$

$$|\beta_{01}\rangle \text{ or } |\psi^+\rangle = \frac{|01\rangle+|10\rangle}{\sqrt{2}}$$

$$|\beta_{10}\rangle \text{ or } |\phi^-\rangle = \frac{|00\rangle-|11\rangle}{\sqrt{2}}$$

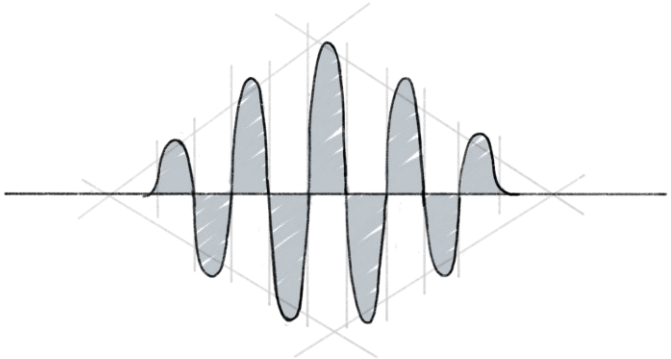
$$|\beta_{00}\rangle \text{ or } |\psi^-\rangle = \frac{|01\rangle-|10\rangle}{\sqrt{2}}$$



https://quantumcomputinguk.org/tutorials/introduction-to-bell-states

Quantum **Interference:**

- occurs when two or more quantum amplitudes (associated with different possible paths or states) combine, leading to an enhancement or suppression of the total probability of an outcome due to their relative phase.
- Due to the wave-like nature of quantum particles



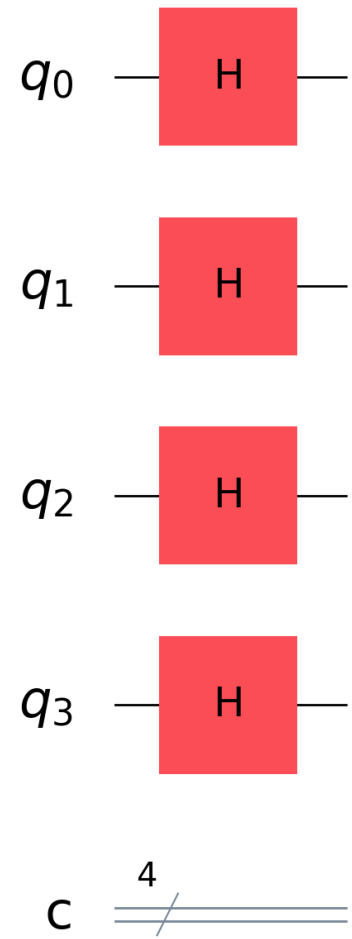
Quantum **parallelism**:

- ***Hadamard transform*** or the *Walsh-Hadamard transform*

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x}^{2^n-1} |x\rangle$$

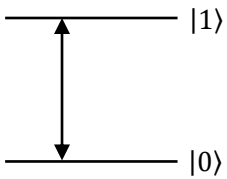
e.g. $H^{\otimes 2}\,|0\rangle^{\otimes 2} = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)$

$$= \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2}}$$

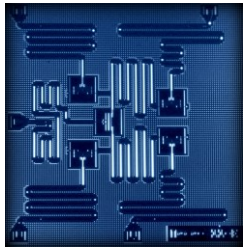


Qubit technology

Theoretically, any two-level quantum system can be used to make a qubit:

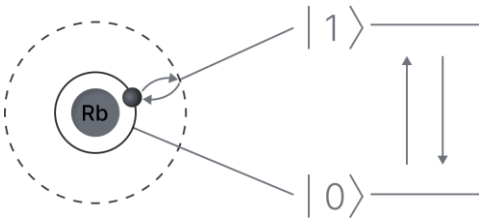


Superconducting qubits



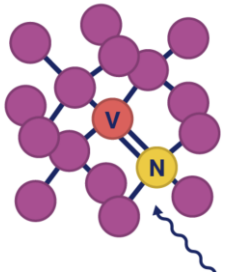
Source: IBM Quantum

Neutral atoms



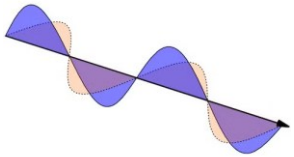
Source: QuEra

Nitrogen vacancy center qubits



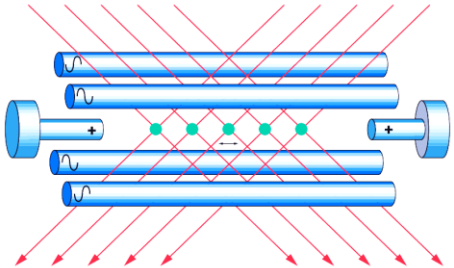
Source: IDTechEx

Photons



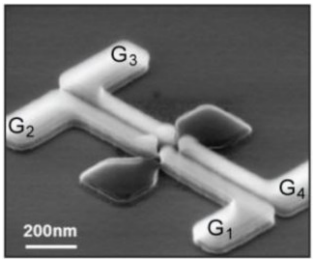
Source: University of Waterloo

Trapped ion qubits



Source: Steane & Rieffel

Quantum dot qubits



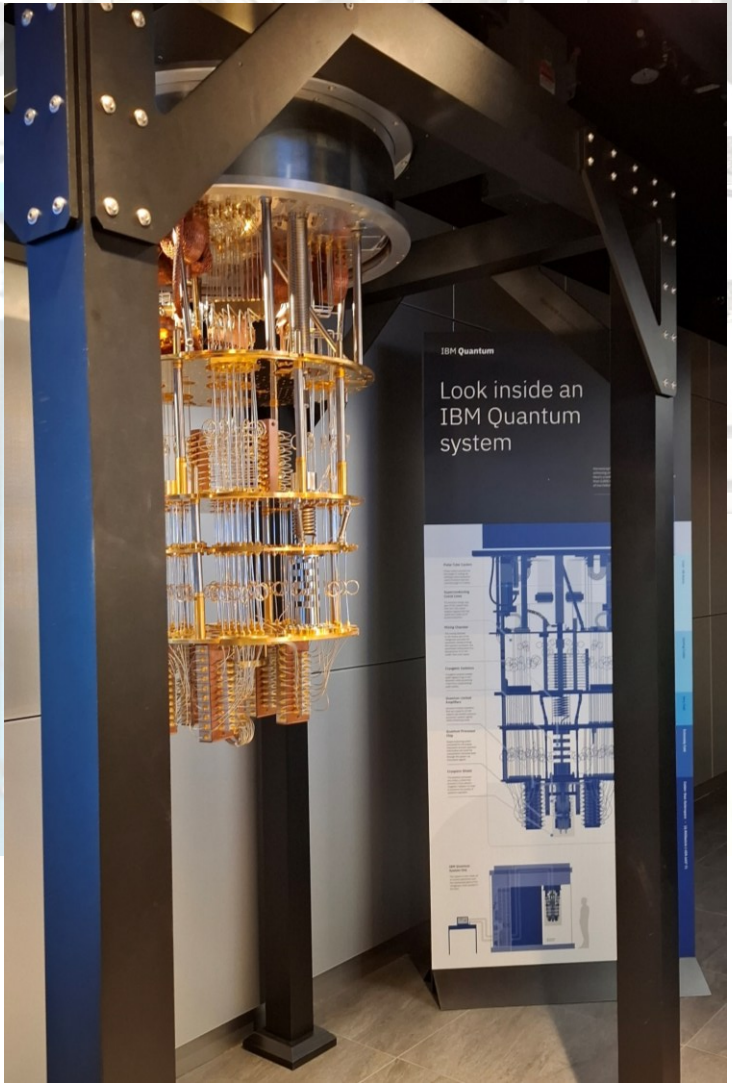
Source: Ansaloni, F

NISQ devices:

- **Noisy Intermediate-Scale Quantum (NISQ)** processors:

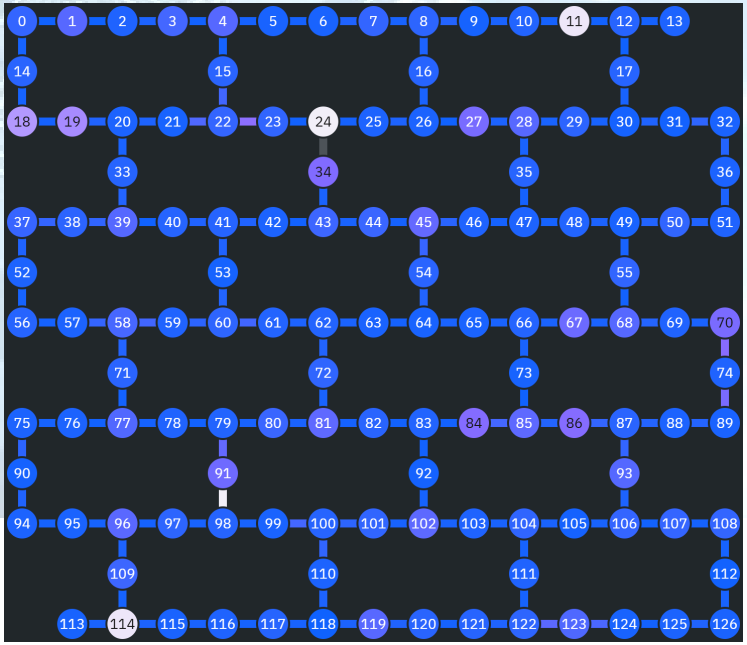
  - **noise** in quantum gates
  - **limits the size** of quantum circuits that can be executed reliably.
  - **imperfect qubits**

  - **intermediate-scale**: in terms of the **number of qubits**



[24] J. Preskill, "Quantum computing in the NISQ era and beyond," Quantum, vol. 2, Aug. 2018, Art. no. 79, doi: 10.22331/q-2018-08-06-79

An IBM quantum processor and the qubit connectivity of an other processor



Quantum noise and error

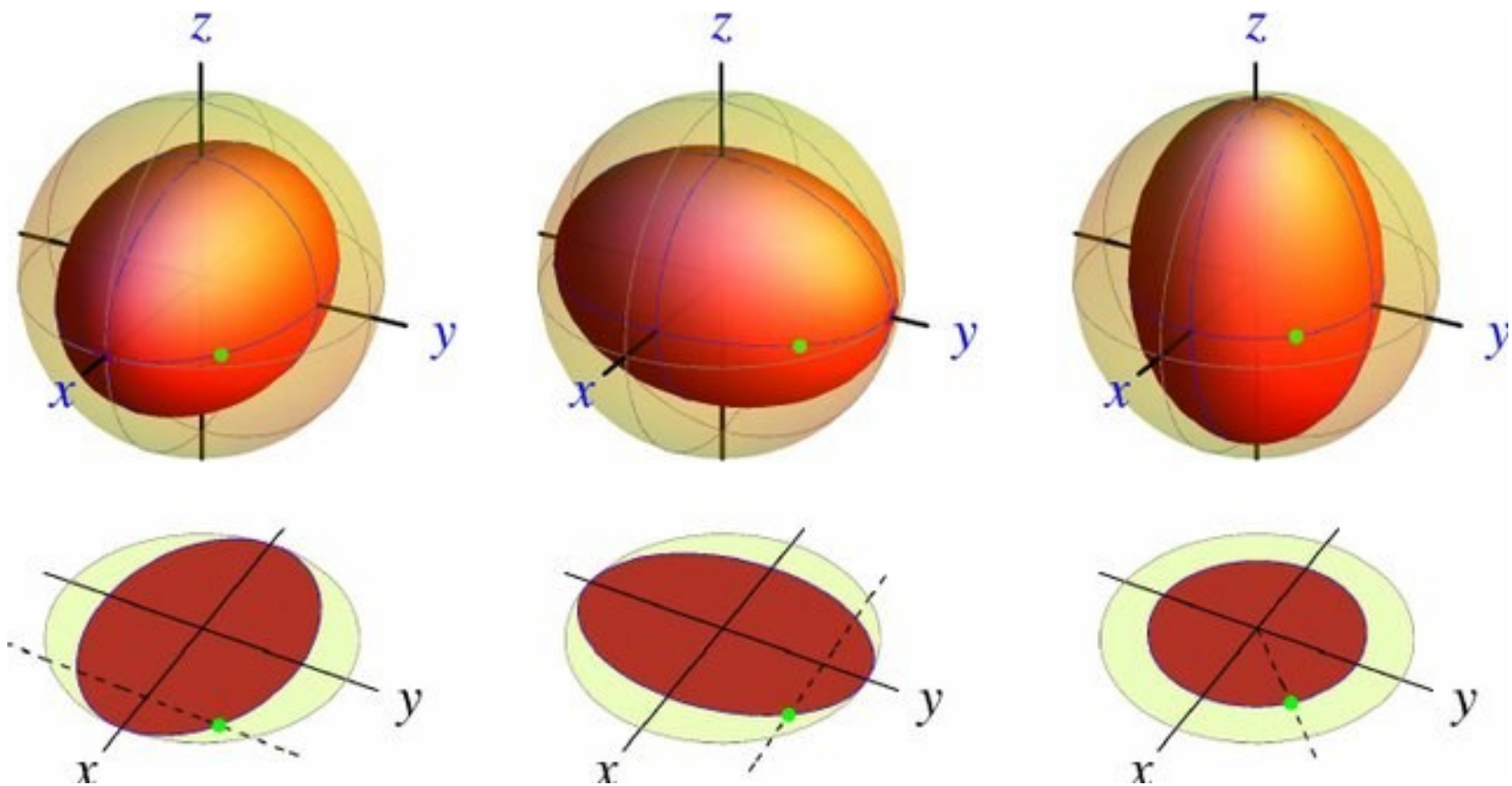
Quantum decoherence: the process in which quantum particles and systems can decay, collapse or change, converting into single states measurable by classical physics.

- Intentionally triggered by measuring qubits to send results back via bits.
- Unintentionally triggered due to environmental factors – undesirable!

Noise: anything that affects our quantum system in an undesirable way.

Source of noise	What is it?
Coherent noise	Imperfect quantum gates causing over- or under-rotations, resulting in outcomes different to the desired
Incoherent noise	Loss of quantum properties in a qubit due to interaction with its environment
State preparation and measurement	Imperfections in preparing initial quantum states and measuring quantum states
Projection noise	The non-deterministic nature of quantum computers resulting in fluctuations in measurement outcomes

Quantum noise



Deformation of the Bloch sphere
- Bit flip, phase flip, both

https://www.researchgate.net/publication/266751221_Optimized_qubit_phase_estimation_in_noisy_quantum_channels

Qiskit

- Provides tools for creating and manipulating quantum programs and running them on prototype quantum devices on IBM Quantum Experience or on simulators on a local computer.

https://en.wikipedia.org/wiki/Qiskit

What is Qiskit?

- The name "Qiskit" is a general term referring to a collection of software for executing programs on quantum computers.

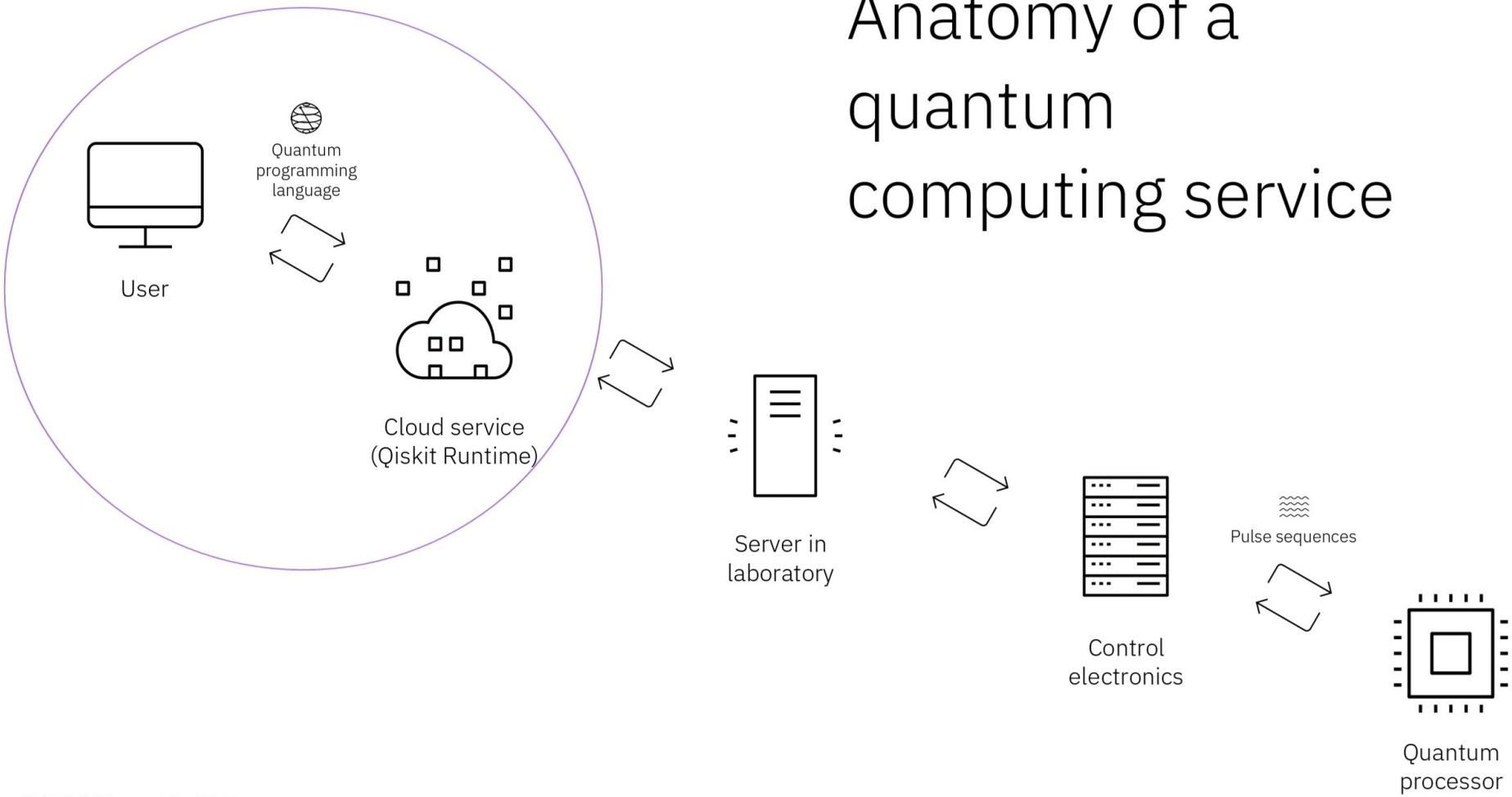
- **Qiskit SDK** is an open-source SDK for working with quantum computers at the level of extended quantum circuits, operators, and primitives.

- github.com/Qiskit/qiskit
- quantum.cloud.ibm.com/docs

- **Qiskit Runtime** is a cloud service to execute workloads on IBM quantum processing units (QPUs).
- The **Qiskit Ecosystem** is a collection of software and projects that build on or extend Qiskit.

qiskit.github.io/ecosystem

Anatomy of a quantum computing service



Qiskit Global Summer School 2024

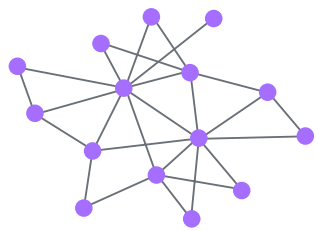
Qiskit 2.0

Released April 2025

- Qiskit 2.0 is the first release to introduce a C API beyond the usual Python interface, marking the beginning of multi-language support - sets the foundation for accessing Qiskit from C directly and building bindings for other programming languages on top of C interface.

- **Two modes of operation:**
- Standalone C usage (no Python runtime dependency) - Installation Guide
- Embedded extension, for calling the C API from a Python context Python - Extension Guide
- Currently, the C interface is minimal but growing: C API Reference
- **Other Qiskit 2.0 highlights:**
- Rust-powered performance boosts: Up to 2× faster circuit construction and ~20% faster transpilation.
- Support for abstract/deferred timing in dynamic circuits: Enables use of stretches for complex timing logic.
- New 'Box' support: Atomic blocks for the transpiler; useful for twirling, noise learning, and more.

Qiskit Pattern:
The anatomy of a quantum algorithm



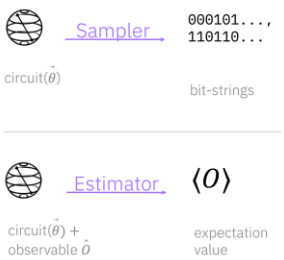
01

Map problem instance to quantum circuits and operators

```
PassManager([UnitarySynthesis(),
             BasisTranslator(),
             EnlargeWithAncilla(),
             AISwap(),
             Collect1qRuns(),
             Optimize1qGates(),
             Collect2qBlocks(),
             ConsolidateBlocks()])
```

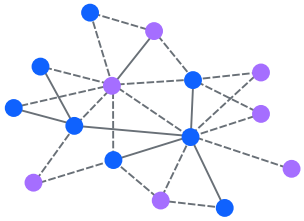
02

Optimize for target hardware execution



03

Execute via Qiskit Runtime



04

Result processing

Map

Optimize

Execute

Post-Process

Qiskit SDK sets the foundation

Qiskit SDK gives us a base layer of building blocks for building and running quantum algorithms

Qiskit Circuit Library

Input:
Domain inputs

Output:
Circuits, observable

Map

Transpiler

Input:
Circuits, observable

Output:
ISA circuit, observable

Optimize

Primitives

Input:
ISA circuit, observable

Output:
Expectation value/samples

Execute

Quantum Info

Input:
Expectation value/samples

Output:
Data objects/visualizations

Post-Process

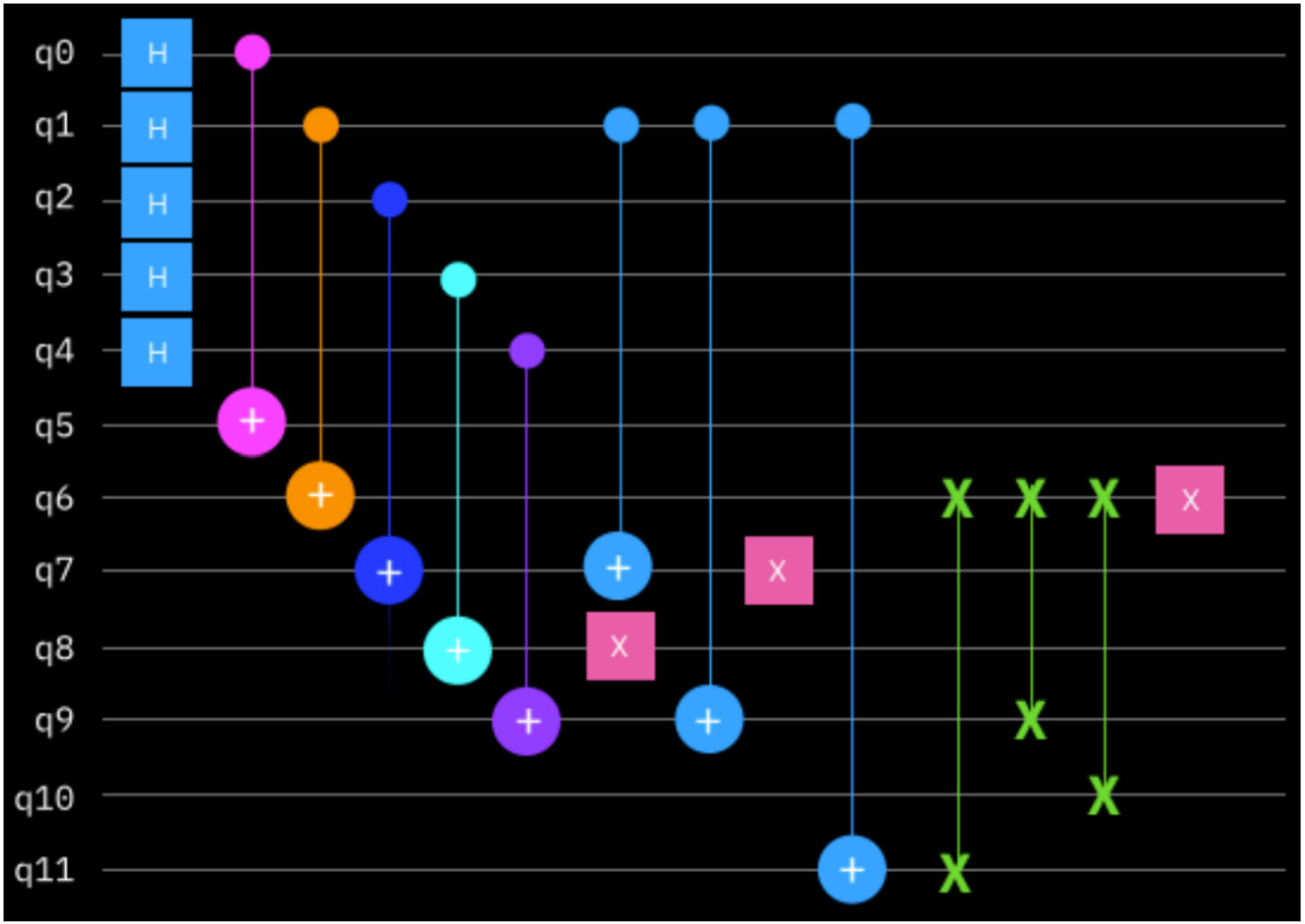
Qiskit addons build on the Qiskit SDK

A collection of research capabilities developed as modular tools that can plug into a workflow to design new algorithms at the utility scale

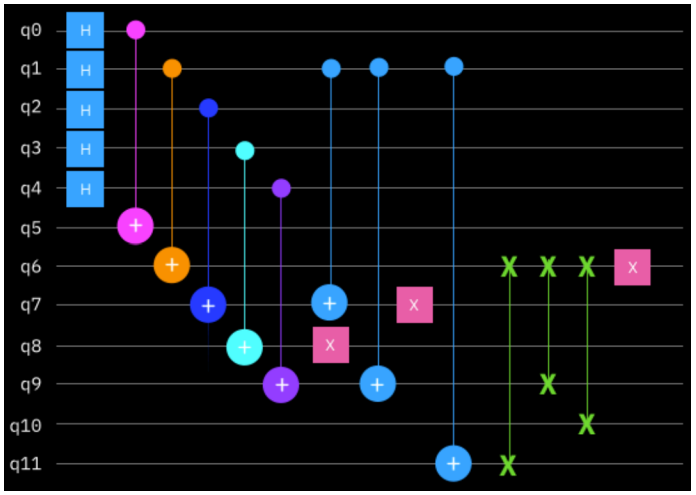
Starting with multi-product formulas (MPF), approximate quantum compilation (AQC-Tensor), operator backpropagation (OBP), and sample-based quantum diagonalization (SQD).

Map	Optimize	Execute	Post-Process
AQC-Tensor	OBP		SQD
MPF	Circuit cutting		M3
Qiskit Circuit Library	Transpiler	Primitives	Quantum Info
Input: Domain inputs	Input: Circuits, observable	Input: ISA circuit, observable	Input: Expectation value/samples
Output: Circuits, observable	Output: ISA circuit, observable	Output: Expectation value/samples	Output: Data objects/visualizations

Circuit depth



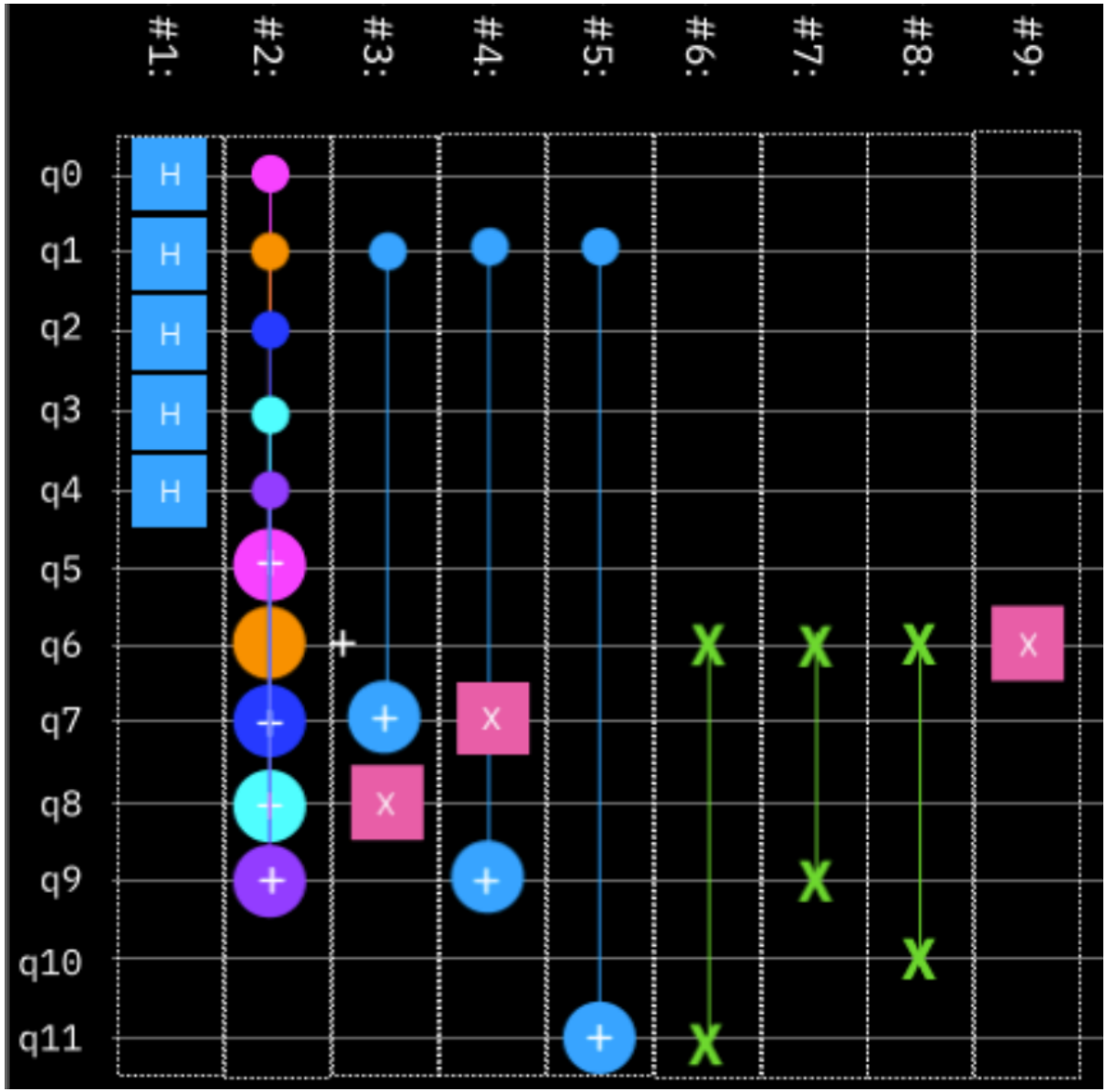
- mutually commuting gates → can be combined into a single layer



Circuit depth
= No. of layers

- all gates are finally transpiled to series of single- and two-qubit gates native to the device

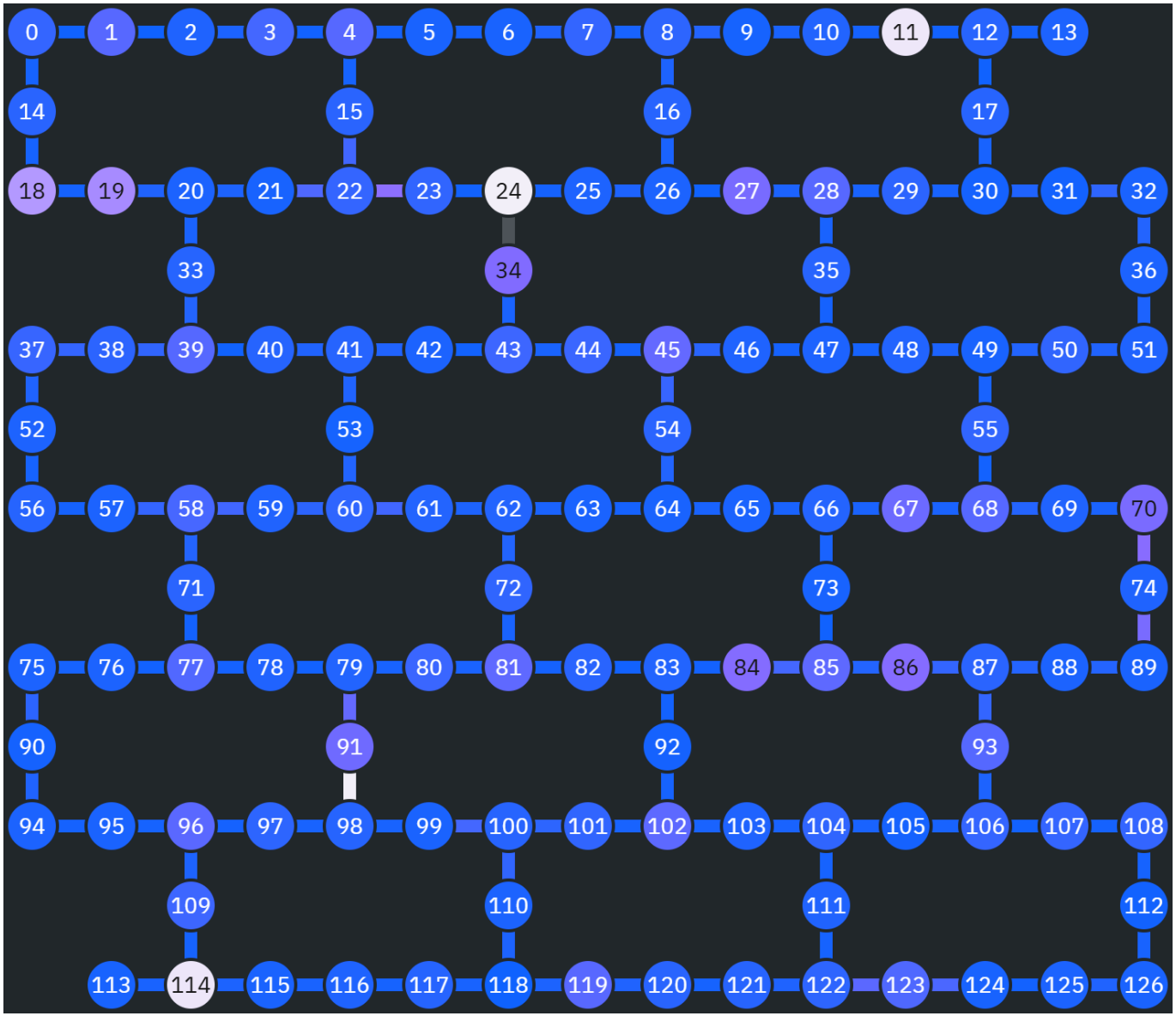
- 2-qubit operations are more costly



https://docs.quantum.ibm.com/api/qiskit/1.0/circuit

- Not nearest-neighbours ⇒ SWAP gates

- Error mitigation and suppression techniques



https://quantum.ibm.com/

Figures of merit (for the IBM devices):

- **Quantum Volume (QV)** → ***quality***:

For a model circuit having *m* qubits and depth *d*, the quantum volume, V_Q can be defined as

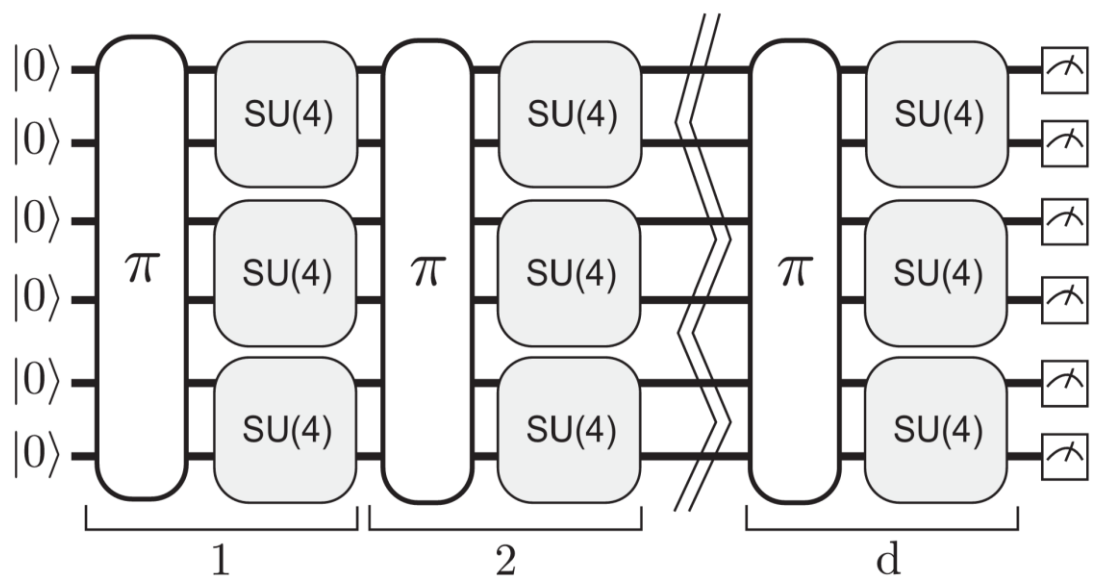
$$V_Q = 2^{\min(d,m)}$$

It quantifies the **largest random circuit** of **equal width and depth** that the computer can successfully implement.

- **Number of qubits** → ***scale***

- ***CLOPS*** → ***speed***

(Circuit Layer Operations Per Second)



Qiskit Installation Instructions

- To Install Qiskit:
- You can use an online jupyter lab environment (quantum.cloud.ibm.com/docs/en/guides/online-lab-environments) or install Qiskit locally.

- Follow the Qiskit installation guide (docs.quantum.ibm.com/guides/install-qiskit) to complete the following steps:

- 1. Install python

- 2. Install Qiskit

    - Create a virtual environment and activate it

    - Install qiskit including the visualization extra packages: `pip install qiskit[visualization]`

    - Install qiskit-ibm-runtime: `pip install qiskit-ibm-runtime`

    - Install jupyter: `pip install jupyter`

qisk.it/install

To Access Hardware:

Follow this guide (docs.quantum.ibm.com/guides/setup-channel) to setup an IBM Quantum account

1. Register IBM Quantum account using your institutional email

2. Save token to your local computer